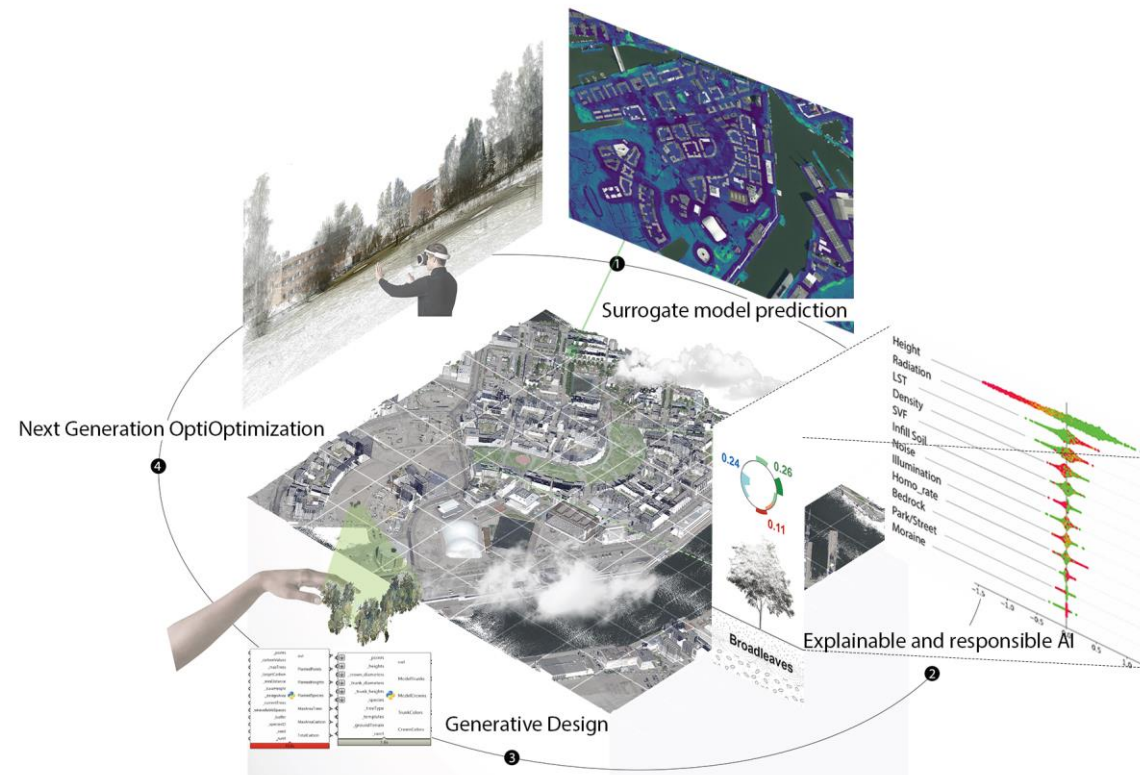


人机共解

预测模型向可解释耦合智能的范式转移

Human-AI Teaming

A Paradigm Shift from Predictive Models to Explainable Coupled Intelligence



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Chapter

1

替代模型简介

**Surrogate Model
Introduction**

2

多维特征提取

**Feature
Extraction**

3

人工智能体训练

**Surrogate Model
Training**

27.06.2026

4

分析模块封装

**Module
Encapsulation**

28.06.2026

5

可解释智能耦合

XAI Coupling

6

基于场景的空间推测

**Scenario
Inference**



Chapter 1

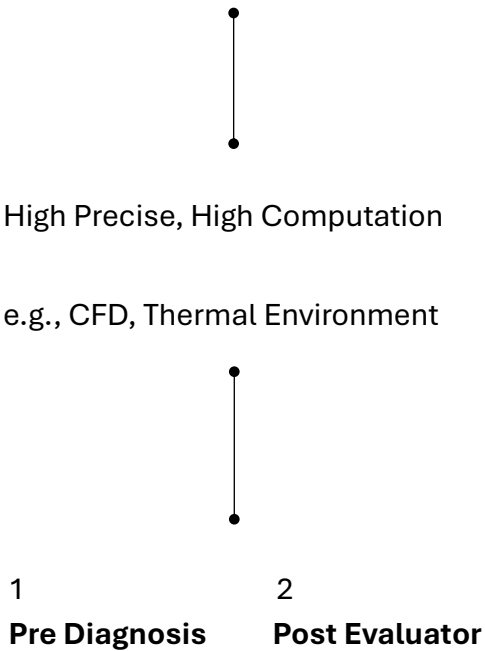
Surrogate Model Introduction: From Simulation to Design Iteration

What is surrogate model

Why surrogate model is necessary

How to design a surrogate model

Traditional Simulation



Traditional Simulation

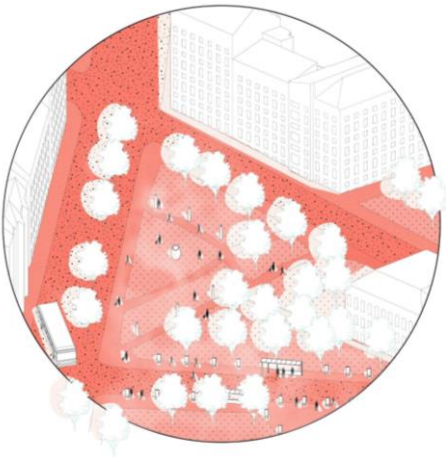
$$y = f(x)$$

Surrogate Model

$$f(x) \approx y$$



Thermal Environment Simulation:



Input

Urban geometry,
Material,
Vegetation,
Meteorological data,

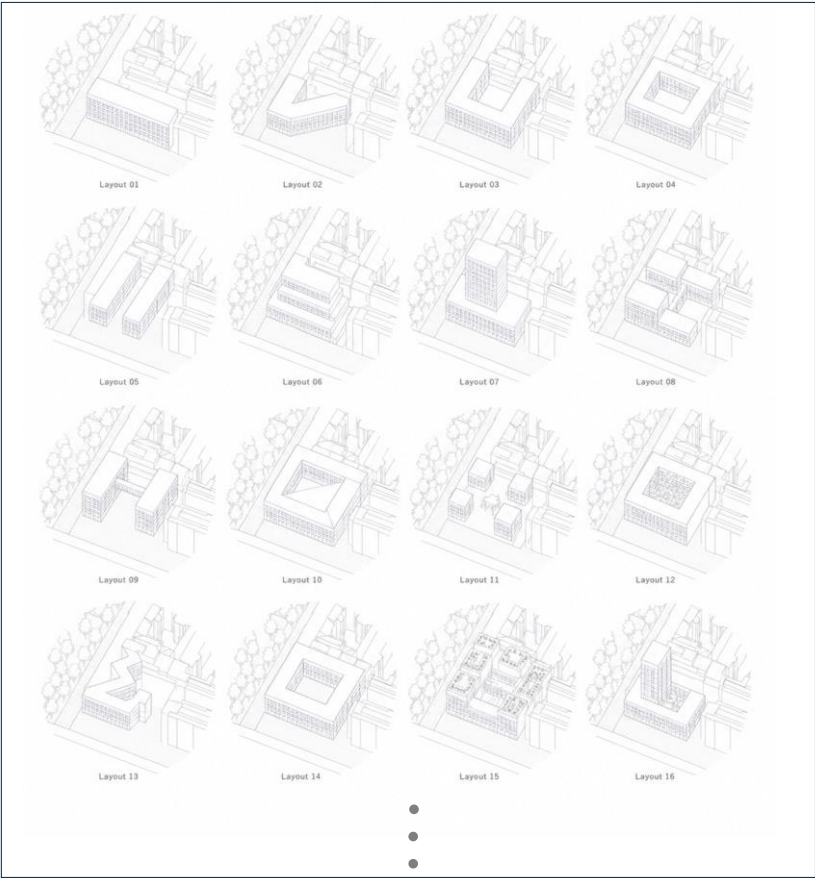
Envi-Met/Ladybug/...

Air Temperature,
Mean Radiant Temperature,
Wind Speed,
Water Vapour Pressure,

UTCI

Universal Thermal Climate Index

$$UTCI = Ta + \text{Offset}(Ta; Tmrt; va; vp)$$
 (K. Blazejczyk, 2013)



Differences

Building height,
Density,
Sky View Factor,
Canyon height/width,
Green coverage,
Canopy cover ...

Envi-Met/Ladybug/...

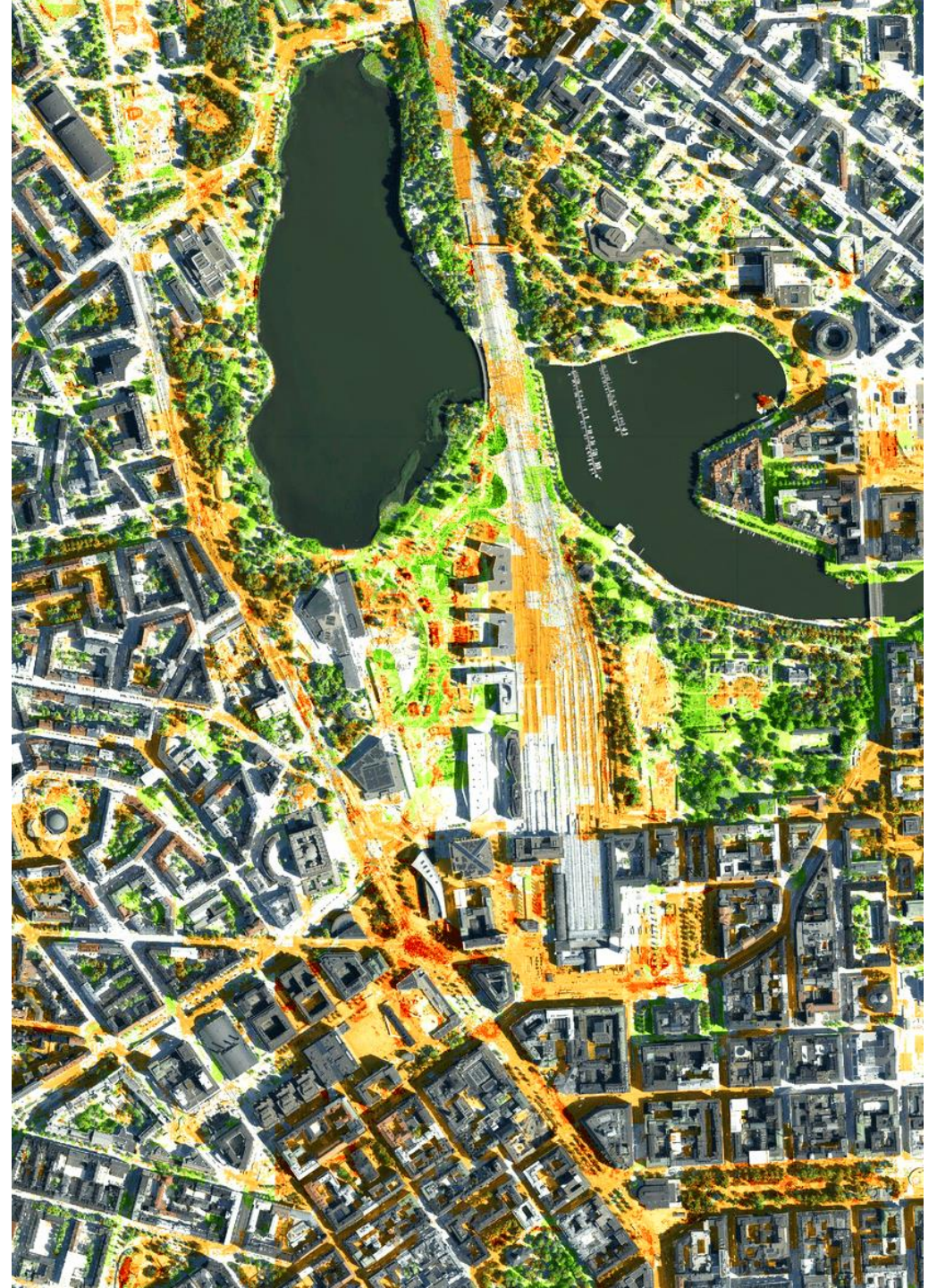
Air Temperature,
Mean Radiant Temperature,
Wind Speed,
Water Vapour Pressure,

Surrogate Model

UTCI
Universal Thermal Climate Index

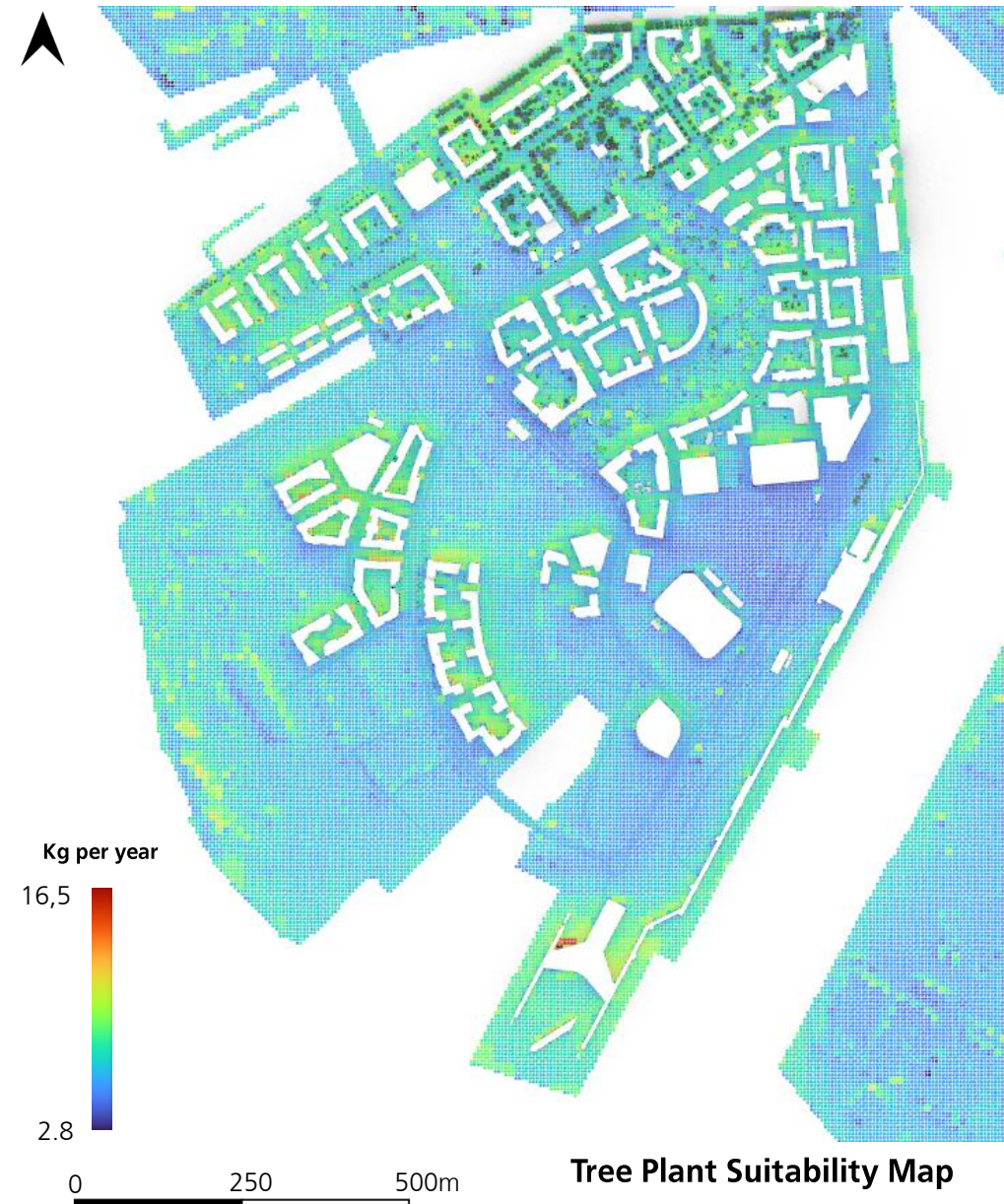
Surrogate Model Bridges the Gaps:

1. Rapid feedback in complex simulations.
2. Allow design indicators for prediction.
3. Understand which spatial variables truly affect the results in design cases.



Surrogate Model Can be Considered:

1. Scalar Prediction – e.g., carbon sequestration, energy consumption, thermal comfort.
2. Spatial Field Prediction – e.g., wind environment, thermal environment, and radiation.
3. Generative Prediction – e.g., floor plan design, urban block arrangement.

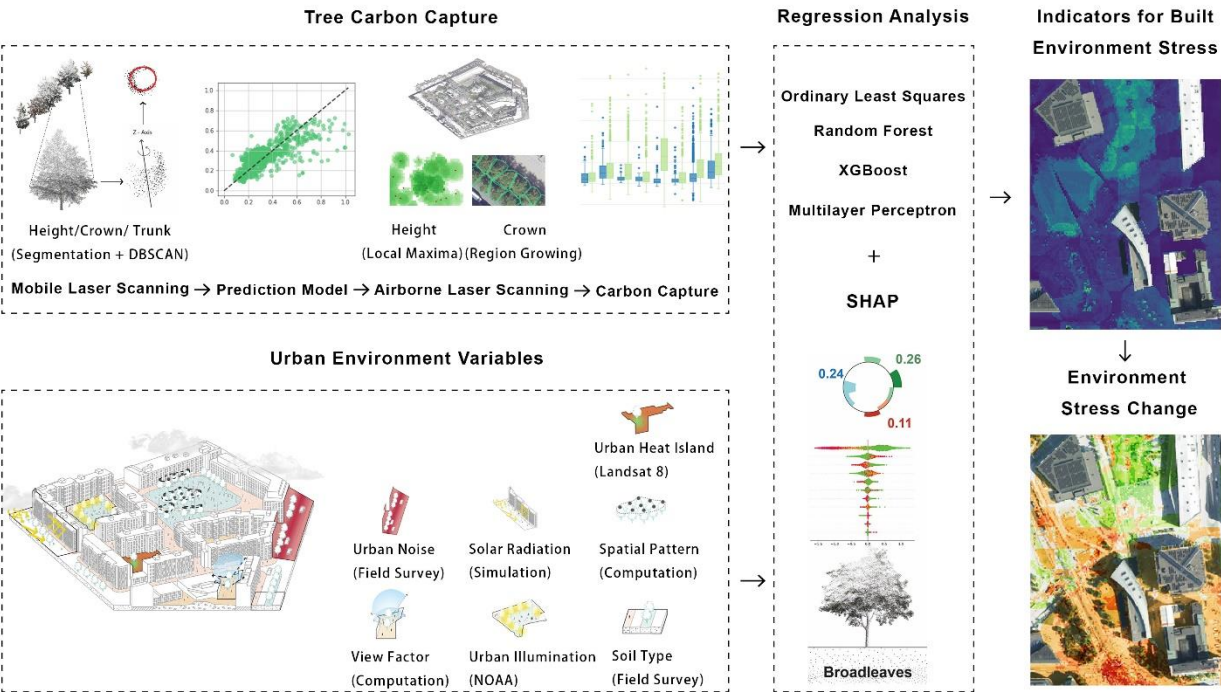


Tree Plant Suitability Map

A colored map visualizing the carbon capture potential per species if plant a tree here.

A Good Surrogate Model Should Include:

- 1. The inputs characterize the differences of design schemes.
- 2. A reasonable clear output.
- 3. Availability of training data.
- 4. Rigorous validation.
- 5. Friendly and Controllable to designer.



Example of a surrogate model development from training to application.

Chapter 2

Feature Extraction: Translate Space into Trainable Data

What kind of features we can translate to variables

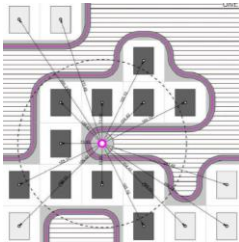
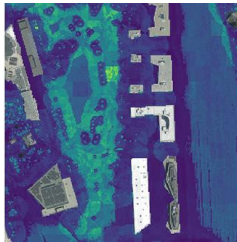
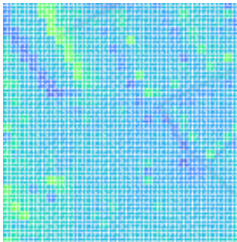
What is the common analysis unit

How to clean data

Feature Types

1	2	3	4	5
Geometry	Environment	Semantic	Topology	Image based
Height	Radiation	Policy related	Distance	Raster image
Density	Wind	Functional term	Connectivity	Rendering
Sky view factor	Temperature	Species	Centrality	Segmentation mask
H/W ratio	Humidity	Type	Nearest Neighbor	Chip (e.g, 4CNN...)

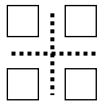
Data Unit for Various Scale



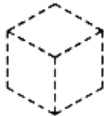
Point



Grid



Block



Design Schemes



Map



Prepare Training Data

Input Features: X

Height (m)	Density (-)	SVF (-)	Green (%)	Wind (m/s)	Radiation (W/m²)	...
28.5	0.72	0.42	31.8	1.24	452.1	...
16.3	0.51	0.58	12.5	1.81	621.3	...
7.2	0.18	0.69	68.4	0.92	318.7	...
22.1	0.64	0.47	16.0	1.37	489.6	...
...	...	...	...	...	...	

Target: Y

Target Value
UTCI (°C)
Carbon (kgCO ₂ e)
Wind (m/s)
Comfort Score (0-100)
...



Data Clean

Missing Value

Outlier Data

Encoding

Normalization

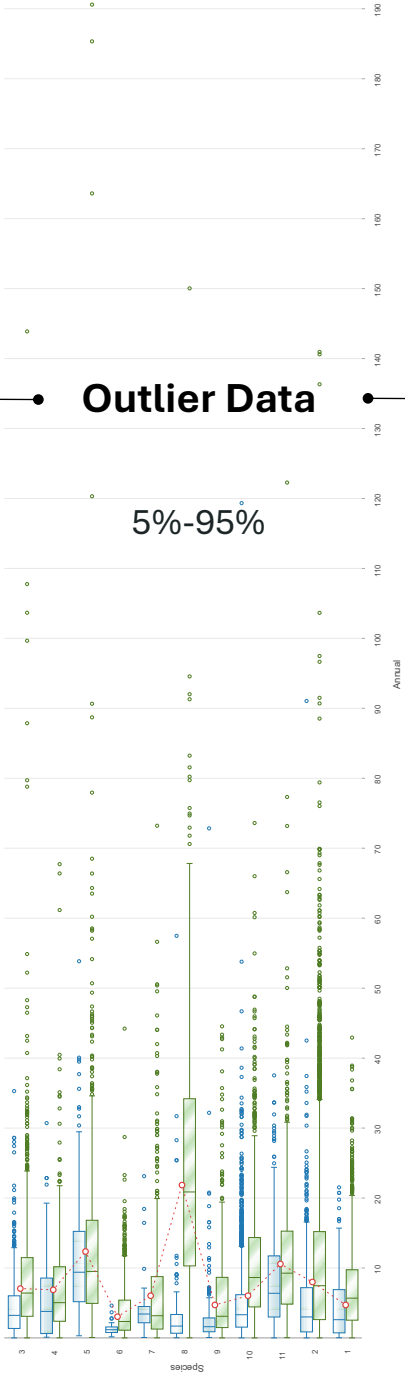
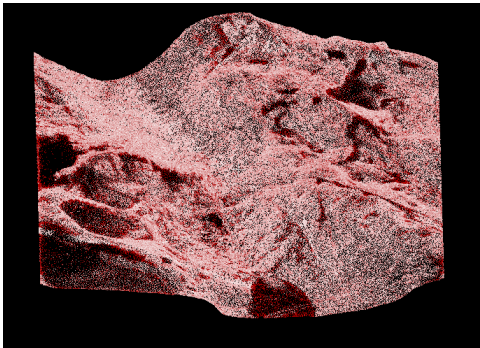
Delete

Interpolate

Neighbor

Specific value

Mosaic



Semantic

Features in different scales

Data from different units/sources

Original Encoding	Ordinal Encoding
Poor	1
Good	2
Very Good	3
Excellent	4

$$X_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \in [0, 1]$$



Chapter 3

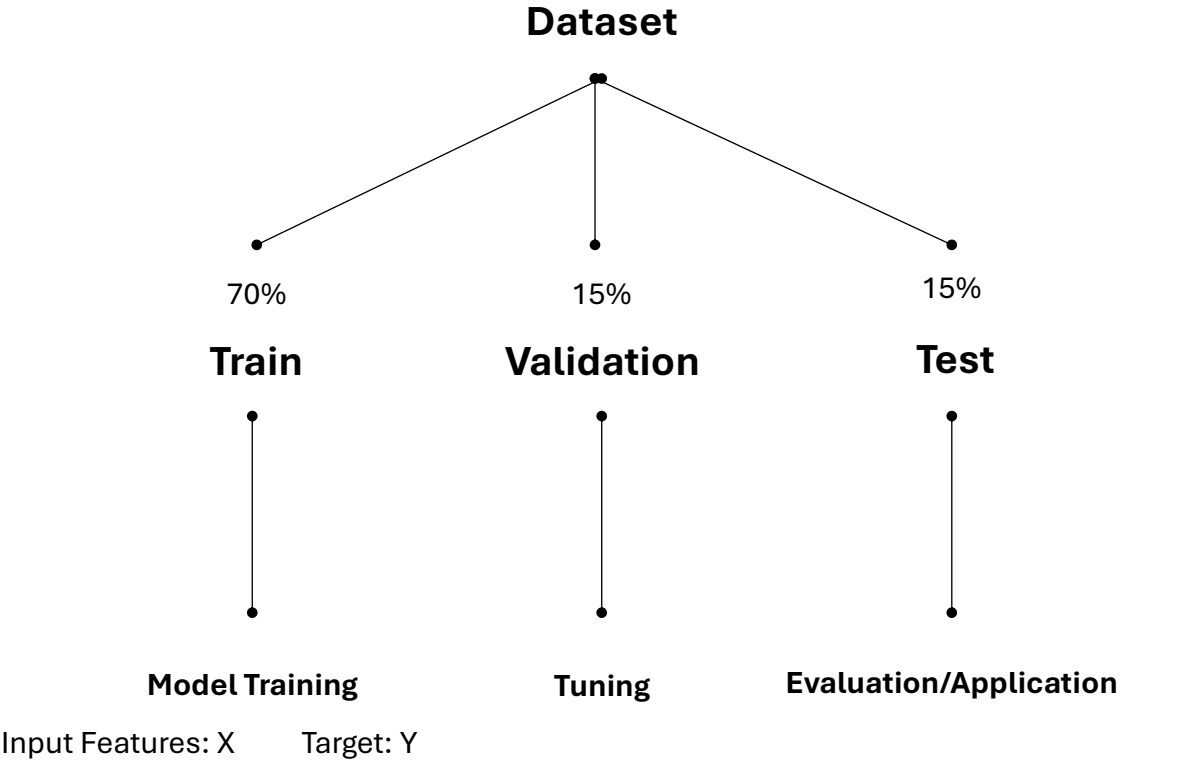
Model Training: Select, Train, and Validate Models

How to select training model

How to select loss to fit

How to tuning parameters

Overall



Model Selection

Linear Model

Ordinary Least Squares

Tree Model

**Random Forest,
XGBoost**

Neural Network

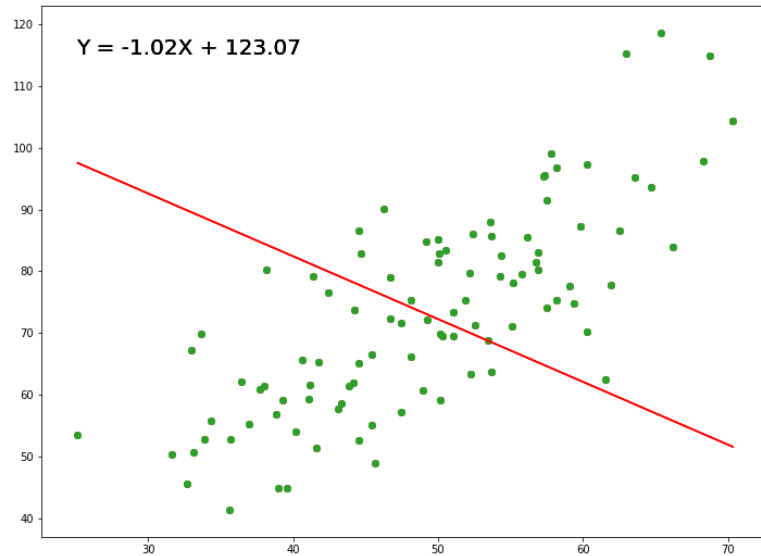
**Multilayer perceptron
Convolution Neural Network**

Generative Network

Generative Adversarial Network

Linear Model: OLS, Ridge, Lasso

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$



Baseline Model

Advantages

Clear coefficient

Fast

Stable

Limits

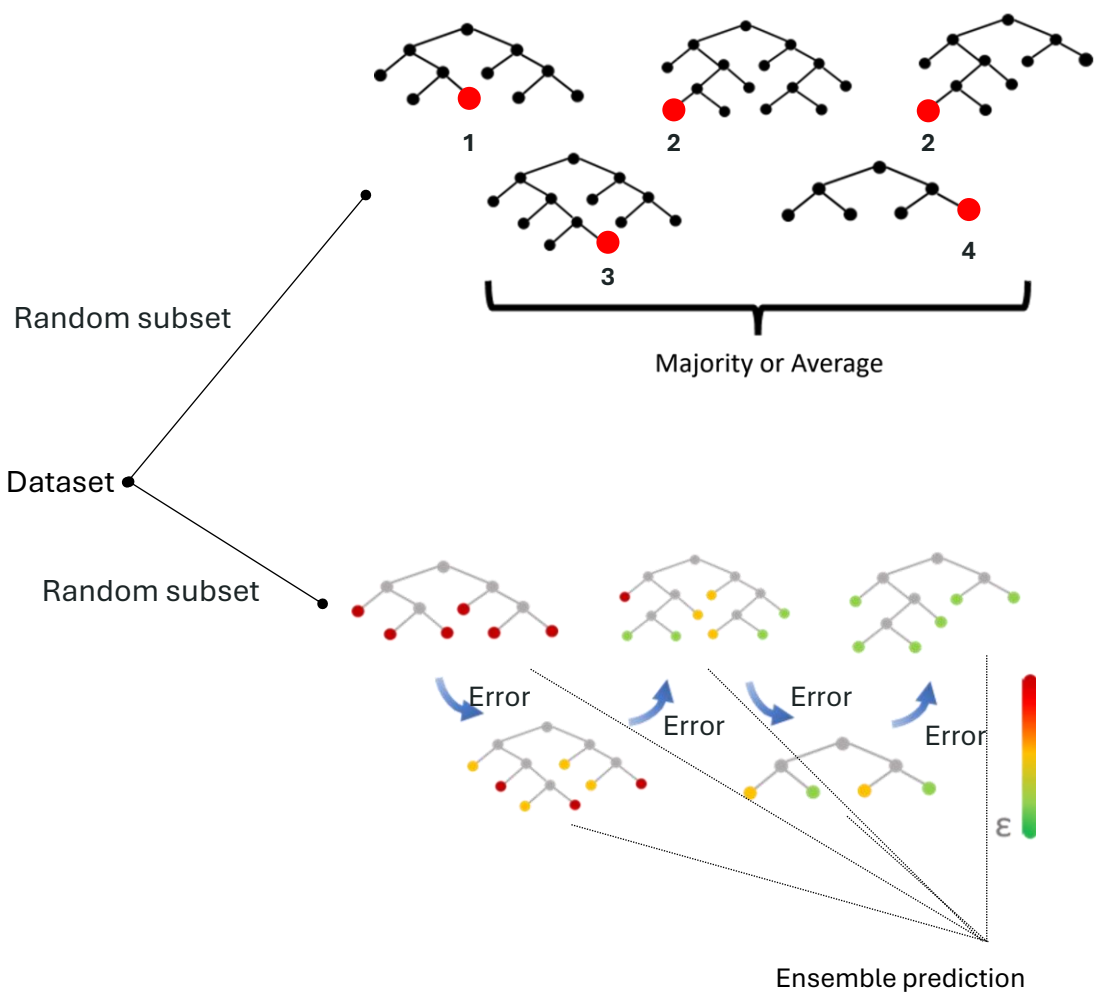
Linear assumption

Weak for interactions

Sensitive for extreme value

Limited features (≤ 8)

Tree Model: RF, XGBoost



Bagging parallel trees

Boosting residual errors

RF

XGBoost

Robust

Higher precision

Low tuning

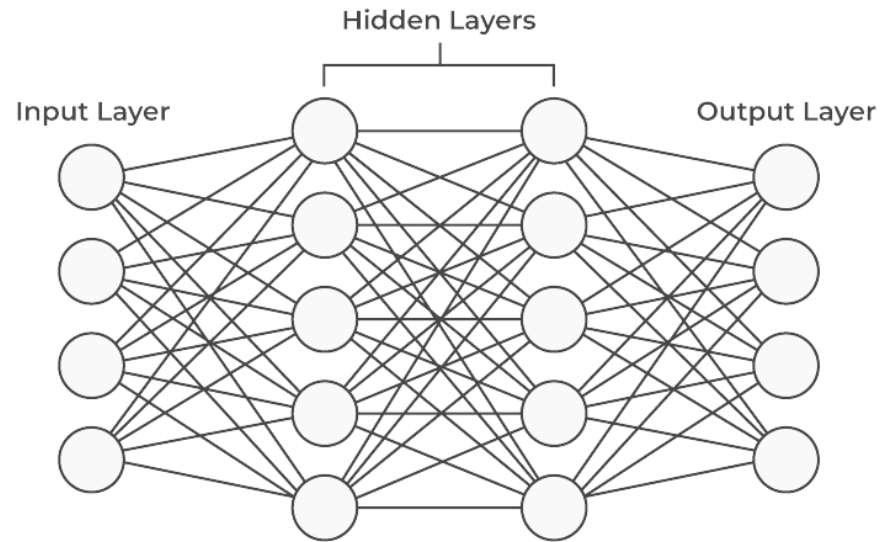
More responsive

Insufficient upper limit

Better in larger dataset

Large size

ANN/MLP



<https://www.geeksforgeeks.org/deep-learning/artificial-neural-networks-and-its-applications/>

Non-linear

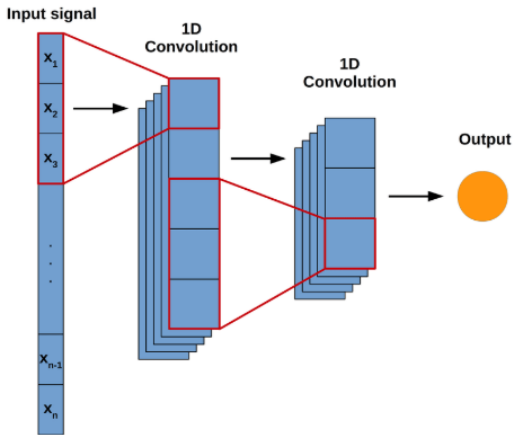
Advantages

- Model complex non-linear features
- Work with many features
- Can learn variable interactions

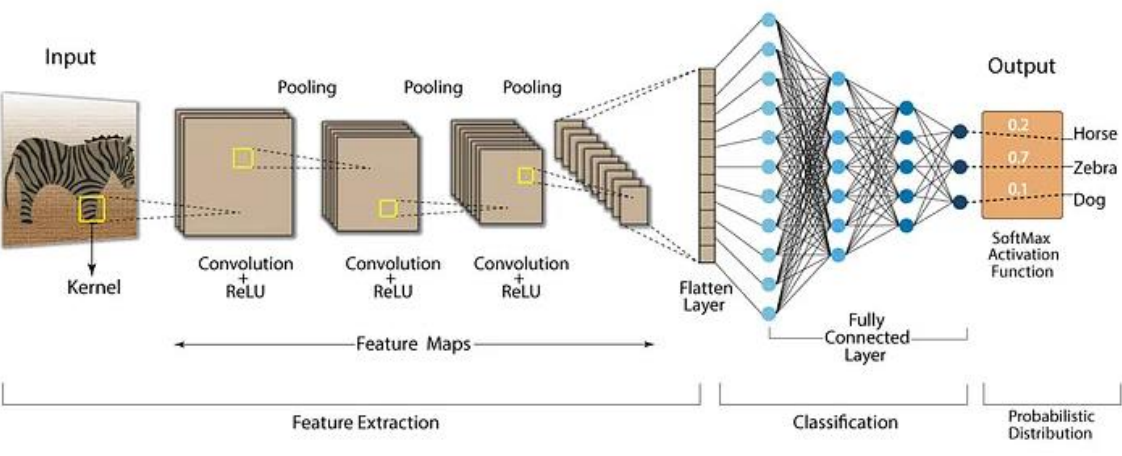
Limits

- Normalization needed
- Large sample needed

CNN



<https://medium.com/@abhishekjainindore24/understanding-the-1d-convolutional-layer-in-deep-learning-7a4cb994c981>



<https://ai.plainenglish.io/deep-dive-into-the-world-of-cnns-8cf22cd84e7>

Advantages

- Recognize local patterns
- Work with images and spatial data

Limits

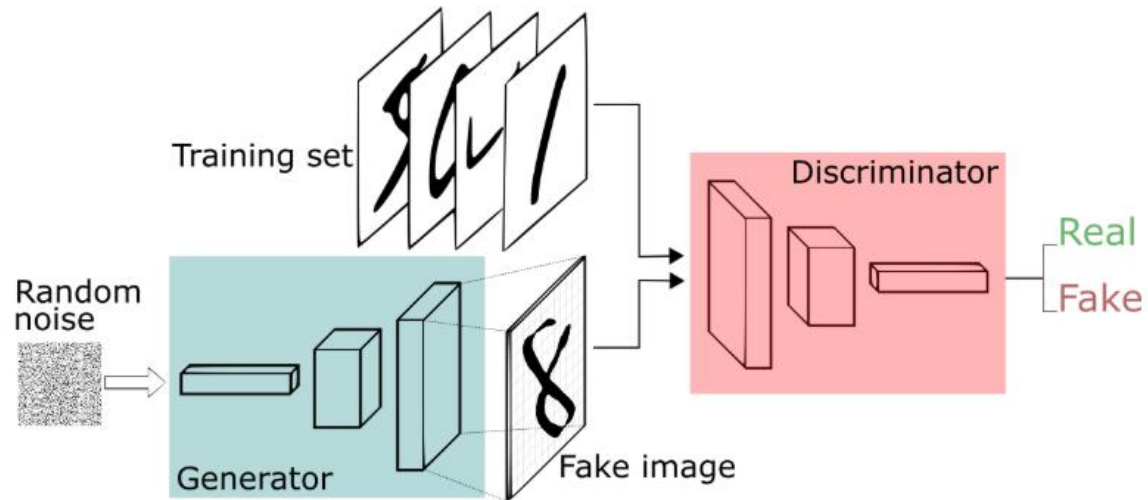
- Large sample needed

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

14	18	22
30	34	38
46	50	54



GAN/PIX2PIX



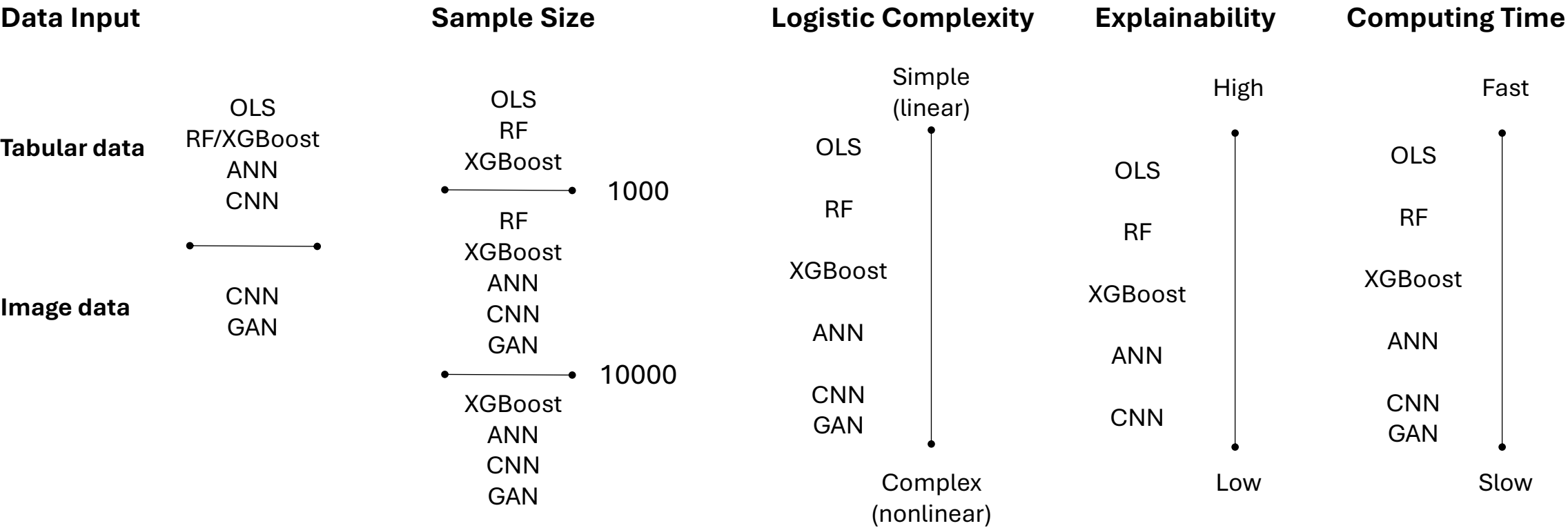
Advantages

- Image to Image
- Work with images and spatial data
- Creative

Limits

- Super uncontrollable
- Large data demand

Model Selection



Validation/Loss

Continuous Regression



Loss

MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Absolute Error

MSE

RMSE

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Root Mean Squared Error

Huber

$$L_{\delta}(a) = \begin{cases} \frac{1}{2} a^2, & \text{if } |a| \leq \delta \\ \delta \left(|a| - \frac{1}{2} \delta \right), & \text{if } |a| > \delta \end{cases}$$

$$a = y_i - \hat{y}_i$$

Quadratic for small errors, linear for large errors

Validation/Loss

Classification



Acer



Alnus



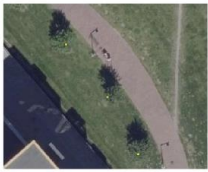
Betula



Malnus



Populus



Prunus



Quercus



Sorbus



Tilia



Ulmus



Picea



Pinus



Tsuga

Loss

Binary Cross Entropy

$$\text{BCE} = - \left[y \log(\hat{p}) + (1 - y) \log(1 - \hat{p}) \right]$$

For binary classification

Categorical Cross Entropy

$$\text{CE} = - \sum_{c=1}^C y_c \log(\hat{p}_c)$$

For multi-class classification

Validation/Loss

Order Categories

Spatial Prediction and Design Suggestion

Environmental Impacts on Trees



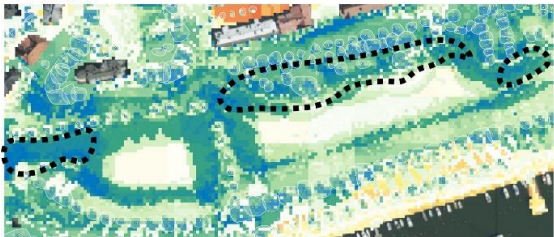
Existing Trees



Alnus



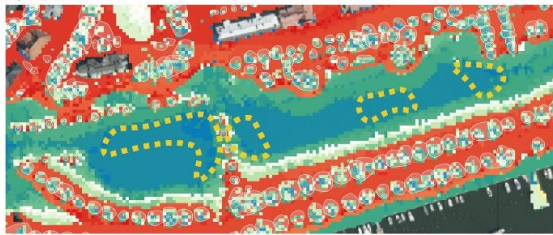
Betula



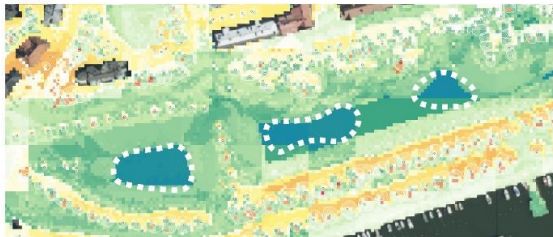
Proposed Desification Plan



Quercus



Tilia



Loss

Ordinal Loss Weighted Cross Entropy

$$L = - \sum_{c=1}^C w_c y_c \log(\hat{p}_c)$$

Useful for ordinal or imbalanced classes

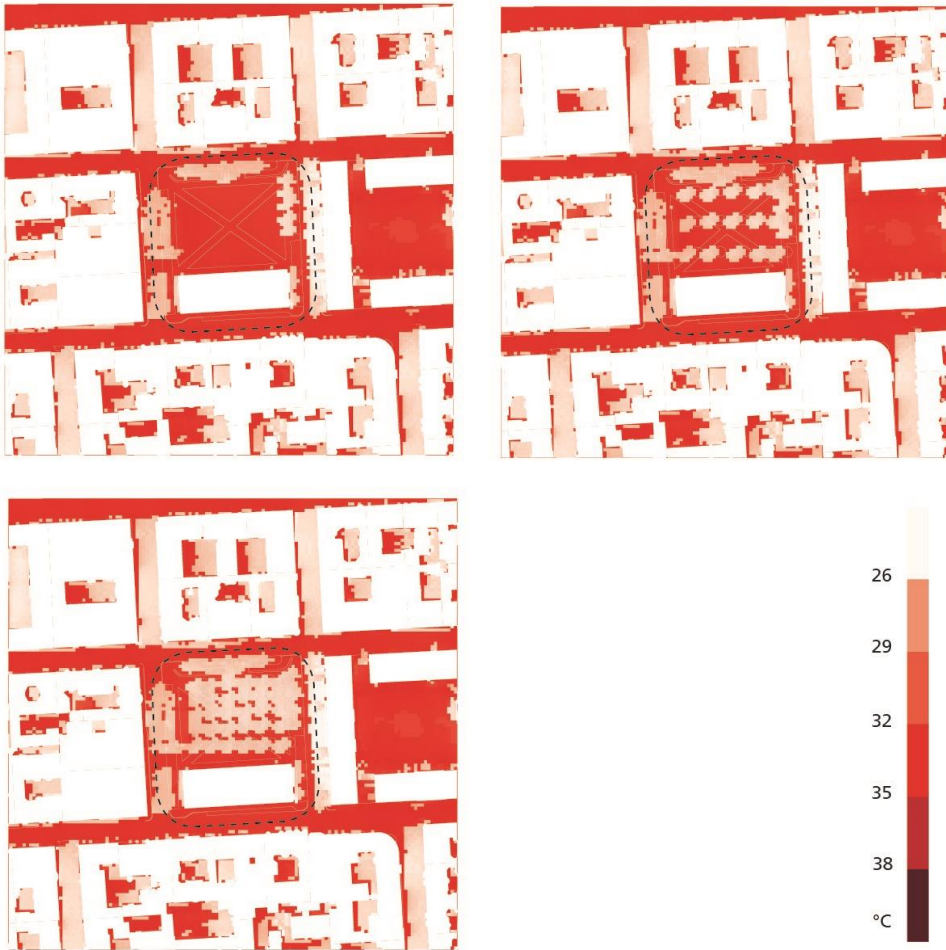
Poisson Loss

$$L = \hat{y} - y \log(\hat{y})$$

Common for count prediction

Validation/Loss

Continuous Values Images



Loss

Pixel-wise L1 Loss

$$L_1 = \frac{1}{N} \sum_{i=1}^N |I_i - \hat{I}_i|$$

Pixel-wise absolute error

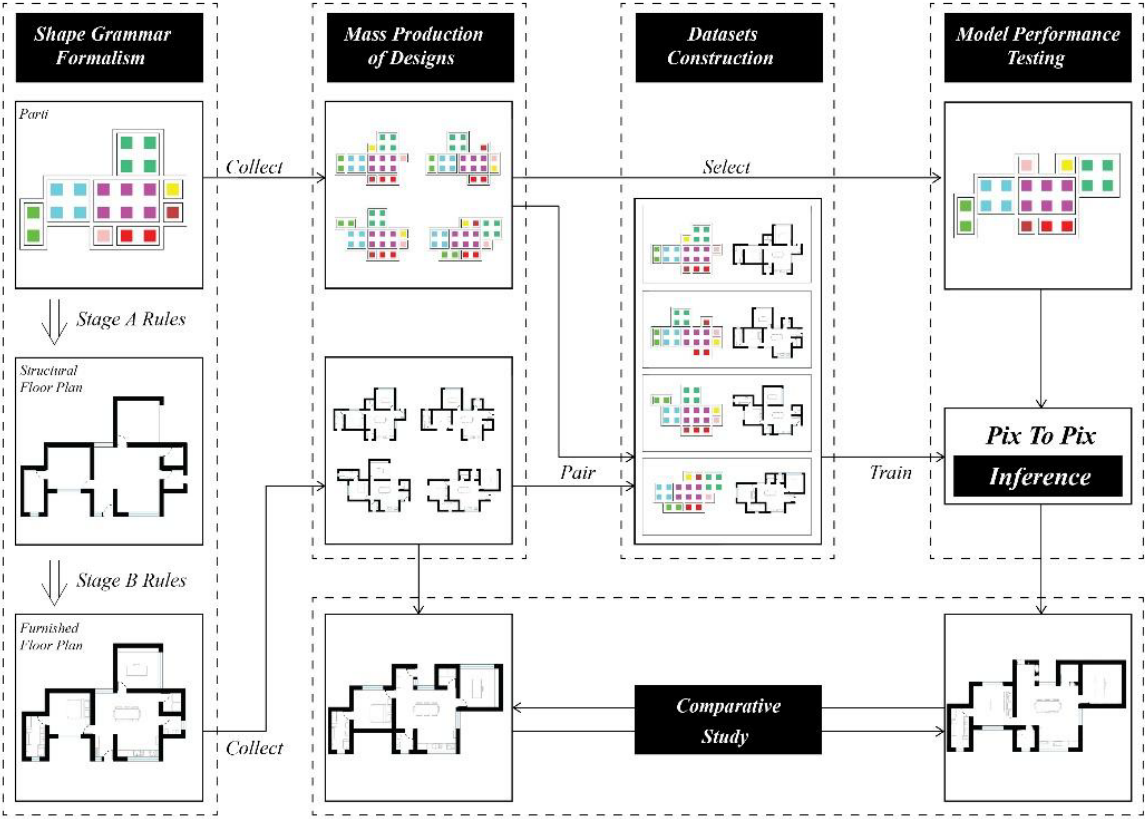
Pixel-wise L2 Loss

$$L_2 = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2$$

Pixel-wise squared error

Validation/Loss

Image Generation



Loss

Discriminator Loss

Generator Loss

$$L_D = - \left[\log D(x) + \log (1 - D(G(z))) \right]$$

GAN discriminator objective

Pixel-wise L1 Loss

$$L_1 = \frac{1}{N} \sum_{i=1}^N |I_i - \hat{I}_i|$$

Pixel-wise absolute error



Validation/Loss

Type	Example	Loss	Focus
Continuous Regression	Temperature:- 10 – 30 oC	MAE, MSE, Huber	Numerical error magnitude
Order Categories	Thermal Comfort: 1(very stress) – 7 (very comfort)	Weighted Cross Entropy, Poison Loss	Reasonable discrete output
Classification	1: Dog 2: Cat 3: Elephant	Cross Entropy	Correct category probability
Continuous Values Images	Thermal Map	L1, L2	Pixels value realism
Images Generation	Design Map	GAN Loss + L1	Visual realism



Model Tuning

Model	Key parameters	Notes
OLS	parameters/features; interaction terms; regularization strength	Baseline and interpretable.
RF	n_estimators; max_depth; max_features; min_samples_split	Robust, less preprocessing.
XGBoost	learning_rate; n_estimators; max_depth; subsample; colsample_bytree; reg_lambda	Powerful but sensitive to tuning.
MLP/ANN	hidden layers; hidden units; learning rate; batch size; dropout / weight decay	Normalize input and watch overfitting.
CNN	number of filters; kernel size; stride / pooling; learning rate; batch size; dropout	Good for image or spatial inputs.



Underfitting / Overfitting

0.01, 0.001, 0.0001

Question?
Thinking?

